Purwadhika Digital Technology School | Digital For Executive

Data Analysis

Session 03:
SQL Group By and Subqueries

Digital for Executive

GROUP BY

Google Search Console
Couverture > Introuvable (404)
Vue d'ensemble
Inspection de l'URL
Performances
Résultats de recherche
Discover
Index
Couverture
Sitemaps
Suppressions
Améliorations
Signaux Web essentiels
Toutes les pages connues
Introuvable (404)
EN SAVOIR PLUS
52
Exemples
MacBook Air

GROUP BY

- The GROUP BY statement groups rows that have the same values into summary rows, like "find the number of customers in each country".
- The GROUP BY statement is often used with aggregate functions (COUNT, MAX, MIN, SUM, AVG) to group the result-set by one or more columns.

```
SELECT column_name(s)
FROM table_name
WHERE condition
GROUP BY column_name(s)
ORDER BY column_name(s);
```

GROUP BY

- Show how many employees in every jobTitle

```
SELECT
   jobTitle, count(employeeNumber)
FROM
   employees
GROUP BY jobTitle;
```

jobTitle	count(employeeNumber)
President	1
VP Sales	1
VP Marketing	1
Sales Manager (APAC)	1
Sale Manager (EMEA)	1
Sales Manager (NA)	1
Sales Rep	17

GROUP BY & WHERE

- Show number of Motorcycles items in each productScale.

```
SELECT
   productScale, COUNT(productCode)
FROM
   products
WHERE productLine = 'Motorcycles'
GROUP BY productScale;
```

productScale	COUNT(productCode)
1:10	3
1:12	1
1:18	2
1:24	3
1:32	3
1:50	1

GROUP BY, WHERE, & AS

- Show number of Motorcycles items in each productScale and rename the columns concisely.

```
SELECT
    productScale, COUNT(productCode) as number_of_motocycles_items
FROM
    products
WHERE productLine = 'Motorcycles'
GROUP BY productScale;
```

productScale	number_of_motocycles_items
1:10	3
1:12	1
1:18	2
1:24	3
1:32	3
1:50	1

GROUP BY, WHERE, AS & HAVING

- Show number of Motorcycles items in each productScale where the number of Motorcycles items above 1 and rename the columns concisely.

```
SELECT
    productScale,
    COUNT(productCode) AS number_of_motocycles_items
FROM
    products
WHERE
    productLine = 'Motorcycles'
GROUP BY productScale
HAVING number_of_motocycles_items > 1;
```

productScale	number_of_motocycles_items
1:10	3
1:18	2
1:24	3
1:32	3

Google Search Console
Couverture > Introuvable (404)
Vue d'ensemble
Inspection de l'URL
Performances
Résultats de recherche
Discover
Index
Couverture
Sitemaps
Suppressions
Améliorations
Signaux Web essentiels
Ergonomie mobile
Logos
Extraits d'avis
Toutes les pages connues
Introuvable (404)
État : Exclues
EN SAVOIR PLUS
Pages concernées
52
Exemples
URL
MacBook Air

Sub-Queries

What is Sub-Queries

Outer Query, executed later

```sql
SELECT
    *
FROM
    payments
WHERE
    amount > (SELECT AVG(amount) FROM payments);
```

Subquery, executed first

- This SQL query is used to show all data which the amount is greater than the average amount.
- Subquery is used to compute the average
- Subquery placed within WHERE clause
- This is only one example!!

Sub-Queries

- A subquery can involve multiple tables.
- For example, we want to access row in table DF_2 which the IDs also available in table DF_1
- We already know the result should be a12, a13, a14

DF_2

var_id	var3
a12	5
a13	6
a14	7
a15	8

```
SELECT
    *
FROM
    df_2
WHERE
    var_id IN (SELECT var_id FROM df_1);
```

DF_1

var_id	var2
a11	1
a11	10
a12	2
a12	4
a13	3
a14	5

Sub-Queries

```sql
SELECT
    *
FROM
    df_2
WHERE
    var_id IN (SELECT var_id FROM df_1);
```

var_id	var3
a12	5
a13	6
a14	7

Sub-Queries

```
SELECT
    *
FROM
    payments
WHERE
    amount > (SELECT AVG(amount) FROM payments);
```

customerNumber	checkNumber	paymentDate	amount
112	HQ55022	2003-06-06	32641.98
112	ND748579	2004-08-20	33347.88
114	GG31455	2003-05-20	45864.03
114	MA765515	2004-12-15	82261.22
114	NR27552	2004-03-10	44894.74
119	LN373447	2004-08-08	47924.19
119	NG94694	2005-02-22	49523.67
121	DB889831	2003-02-16	50218.95
121	MA302151	2004-11-28	34638.14
124	AE215433	2005-03-05	101244.59
124	BG255406	2004-08-28	85410.87
124	ET64396	2005-04-16	83598.04
124	HI366474	2004-12-27	47142.70
124	HR86578	2004-11-02	55639.66

Thank You!

PURWADHIKA HEAD OFFICE

BSD Green Office Park 9 - G Floor,
BSD City, Tangerang, Banten

CAMPUS BRANCHES

Purwadhika BSD
Purwadhika Jakarta
Purwadhika Bandung

CONTACT

021 - 5012 3161
0813 - 1867 8671
0812 - 8060 7160

EMAIL

admission1@purwadhika.com
admission2@purwadhika.com
admission3@purwadhika.com

WEBSITE

www.purwadhika.com